**NurNabi Academy**
IGCSE Mathematics · exam-style revision

# IGCSE Revision Paper

NAME

CLASS / CENTRE

DATE

**MATHEMATICS** **NNA 0580/04**

Paper 4 Calculator (Extended) **October/November 2026**

Full Revision Paper 1 **2 hours**

You must answer on the question paper.

You will need: Geometrical instruments

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name in the box at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- You should use a scientific calculator where appropriate.
- You may use tracing paper.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π, use either your calculator value or 3.142.

**INFORMATION**

- The total mark for this paper is **100**.
- This document has **21** pages.
- The number of marks for each question or part question is shown in brackets [ ].

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**[Turn over**

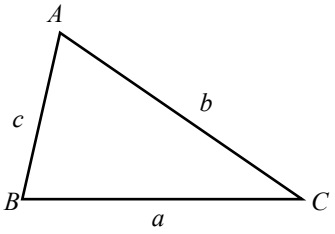
## List of formulas

| | |
|---|---|
| Area, $A$, of triangle, base $b$, height $h$. | $A = \frac{1}{2}bh$ |
| Area, $A$, of circle of radius $r$. | $A = \pi r^2$ |
| Circumference, $C$, of circle of radius $r$. | $C = 2\pi r$ |
| Curved surface area, $A$, of cylinder of radius $r$, height $h$. | $A = 2\pi rh$ |
| Curved surface area, $A$, of cone of radius $r$, sloping edge $l$. | $A = \pi rl$ |
| Surface area, $A$, of sphere of radius $r$. | $A = 4\pi r^2$ |
| Volume, $V$, of prism, cross-sectional area $A$, length $l$. | $V = Al$ |
| Volume, $V$, of pyramid, base area $A$, height $h$. | $V = \frac{1}{3}Ah$ |
| Volume, $V$, of cylinder of radius $r$, height $h$. | $V = \pi r^2 h$ |
| Volume, $V$, of cone of radius $r$, height $h$. | $V = \frac{1}{3}\pi r^2 h$ |
| Volume, $V$, of sphere of radius $r$. | $V = \frac{4}{3}\pi r^3$ |

For the equation $ax^2 + bx + c = 0$, where $a \neq 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For the triangle shown,




$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\text{Area} = \frac{1}{2}ab\sin C$$

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**1** Solve.

$$4(3x - 5) = 2(x + 9) + 2$$

$x =$ ..... [3]

**2** Work out.

$$\frac{16.71 + 46.13}{\sqrt{8.6^2 - 3.5^2}}$$

Give your answer correct to 2 significant figures.

..... [2]

**3** In a sale, the price of a printer is reduced by 18% to \$287.

Work out the original price of the printer.

\$ ..... [2]

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**4** Amira, Bilal and Chen share \$4560 in the ratio 5 : 7 : 12.

Calculate the amount Chen receives.

\$ ..... [2]

**5** These are the first 5 terms of a sequence.

2 6 12 20 30

**(a)** Find the next term of this sequence.

..... [1]

**(b)** Find the $n$th term of this sequence.

..... [2]

**6** Factorise fully.

$6x^2 - 7x - 3$

..... [2]

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**7** Amira invests \$950 at a rate of 3.4% per year compound interest.

Calculate the value of her investment at the end of 6 years.
Give your answer correct to the nearest dollar.

\$ ..... [3]

**8** Bilal invests \$1330 at a rate of $r$% per year **simple** interest.
At the end of 8 years the total value of his investment is \$1968.40.

Calculate the value of $r$.

$r =$ ..... [3]

**9** Solve the simultaneous equations.
You must show all your working.

$3x + 2y = 4$
$5x - 4y = 25$

$x =$ ..... , $y =$ ..... [3]

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**10** Simplify fully.

$$\frac{3x^2 - 12}{x^2 - x - 6}$$

.....

[3]

**11** Make $x$ the subject of this formula.

$$y = \frac{3x + 2}{x - 5}$$

$x =$ .....

[4]

**12** The interior angle of a regular polygon is 162°.

Find the number of sides of this polygon.

.....

[2]

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**13** A curve has equation $y = x^3 - 6x^2 + 9x + 2$.

Find the coordinates of the two turning points of the curve and determine whether each one is a maximum or a minimum.
You must show all your working.

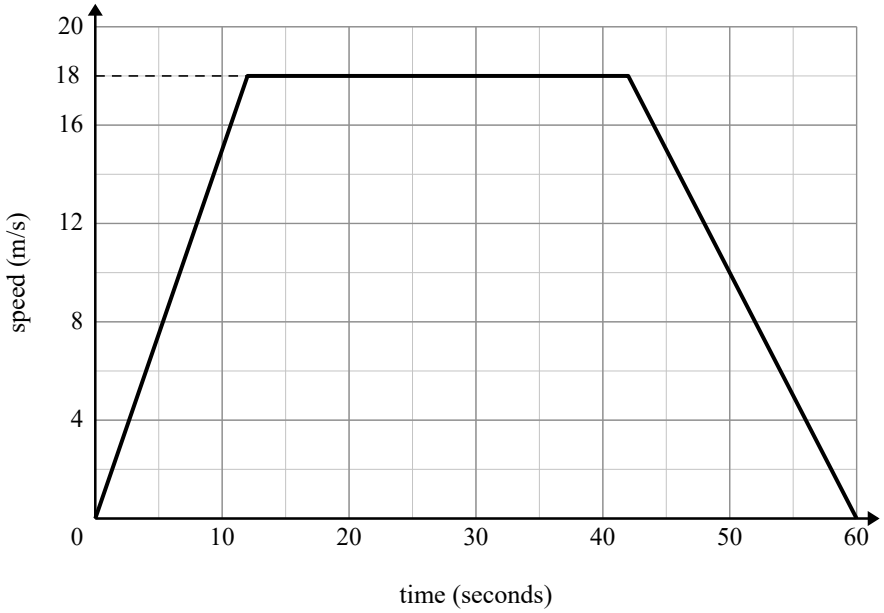
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( _____ , _____ ) _____ [5]

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**14** The speed–time graph shows the journey of a car.




**(a)** Calculate the acceleration during the first 12 seconds.

..... m/s$^2$ [1]

**(b)** Calculate the total distance travelled.

..... m [3]

**(c)** Calculate the average speed for the whole journey.

..... m/s [1]

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**15** A machine is worth \$24 000.
Its value decreases exponentially by 12% each year.

**(a)** Calculate the value of the machine after 5 years.

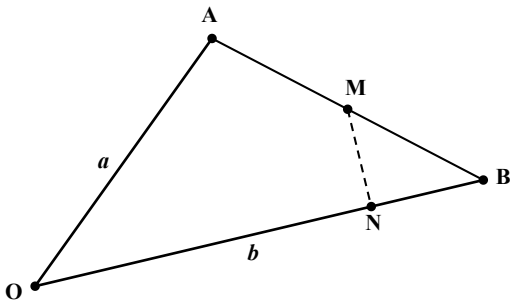
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**(b)** Find the number of complete years before the value of the machine first falls below \$8000.

..... years [2]

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**16** In the diagram, **OA** = $\boldsymbol{a}$ and **OB** = $\boldsymbol{b}$.
M is the midpoint of AB and N is the point on OB such that ON : NB = 3 : 1.




Find, in terms of $\boldsymbol{a}$ and $\boldsymbol{b}$, giving each answer in its simplest form.

**(a) AB**

NOT TO SCALE

..... [1]

**(b) OM**

..... [2]

**(c) MN**

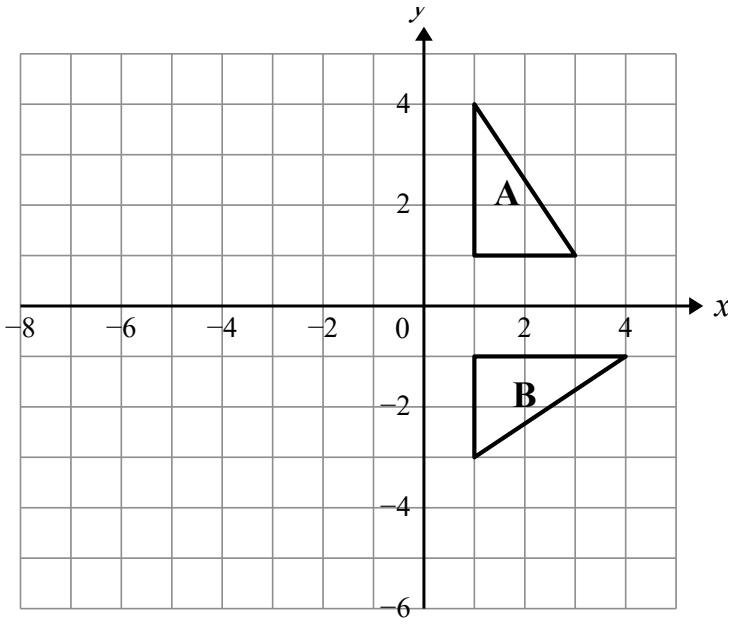
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**17** The diagram shows triangle A and triangle B on a grid.




**(a)** Describe fully the **single** transformation that maps triangle A onto triangle B.

.....

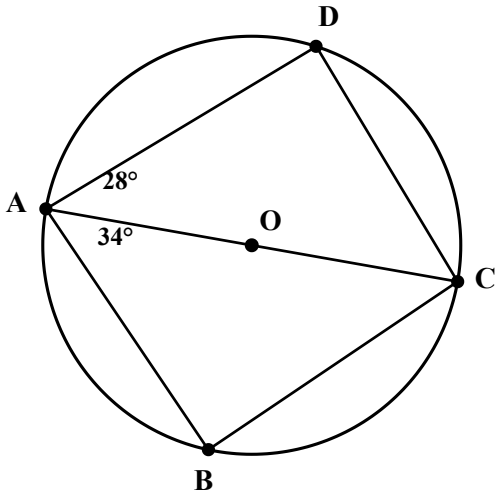
[3]

**(b)** On the grid, draw the image of triangle A after an enlargement with scale factor −2 and centre (0, 1).

[2]

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**18** A, B, C and D lie on a circle, centre O.
AC is a diameter, angle BAC = 34° and angle CAD = 28°.




NOT TO SCALE

Find the following angles, giving a geometrical reason for each answer.

**(a)** Angle ABC

Angle ABC = ..... ° [1]

Reason ..... [1]

**(b)** Angle BCA

Angle BCA = ..... ° [1]

**(c)** Angle BCD

Angle BCD = ..... ° [1]

Reason ..... [1]

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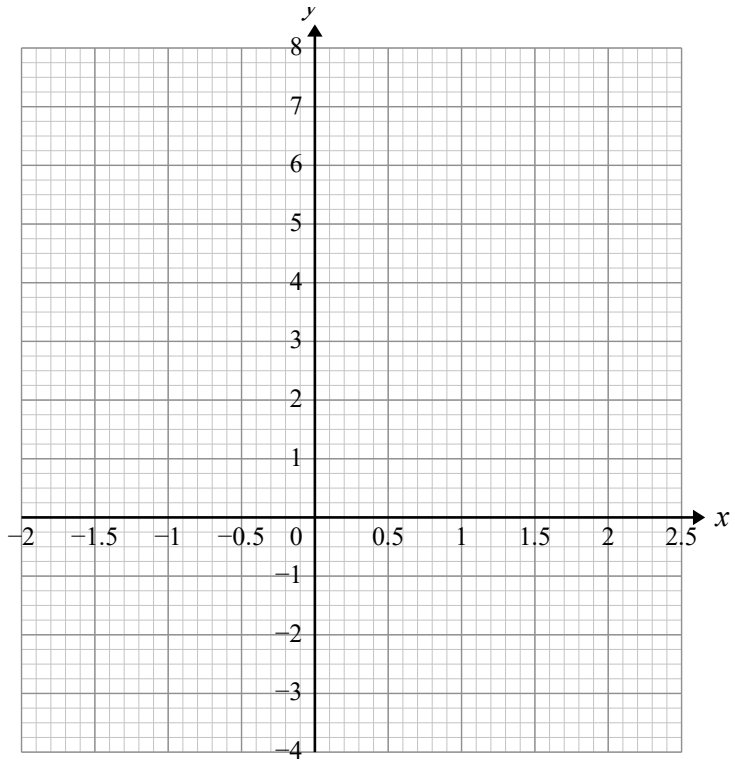
**19** The table shows some values for $y = x^3 - 3x - 1$.

| $x$ | −2 | −1.5 | −1 | −0.5 | 0 | 0.5 | 1 | 1.5 | 2 | 2.5 |
|---|---|---|---|---|---|---|---|---|---|---|
| $y$ | −3 | | 1 | 0.375 | −1 | −2.375 | −3 | | 1 | 7.125 |

**(a)** Complete the table.

[2]

**(b)** On the grid, draw the graph of $y = x^3 - 3x - 1$ for $-2 \leqslant x \leqslant 2.5$.




[4]

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**(c)** By drawing a suitable tangent, estimate the gradient of the graph at the point where $x = 2$.

.....

[3]

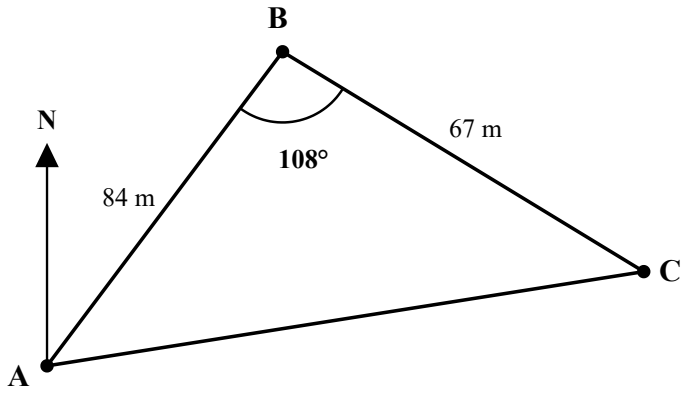
**Question 20 is on page 15.**

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**20** A, B and C are three points on level ground.
AB = 84 m, BC = 67 m and angle ABC = 108°.




NOT TO SCALE

**(a)** Calculate AC.

AC = ..... m [4]

**(b)** Calculate angle BAC.

Angle BAC = ..... ° [3]

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**(c)** The bearing of B from A is 052° and C lies to the east of the line AB.
Find the bearing of C from A.

.....° [1]

**(d)** A vertical mast of height 30 m stands at C.
Calculate the angle of elevation of the top of the mast from A.

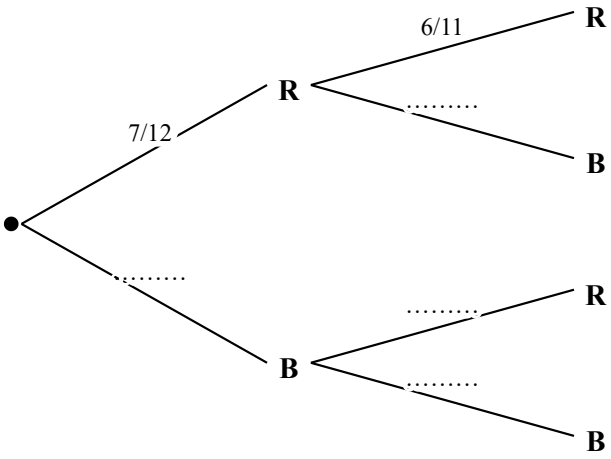
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**21** A bag contains 7 red counters and 5 blue counters.
Two counters are taken at random from the bag, without replacement.

**(a)** Complete the tree diagram.




[2]

**(b)** Find the probability that both counters are red.

.....

[2]

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**(c)** Find the probability that the two counters are different colours.

.....

[2]

**(d)** Three counters are now taken at random from the original bag, without replacement.
Find the probability that at least one of these counters is blue.

.....

[3]

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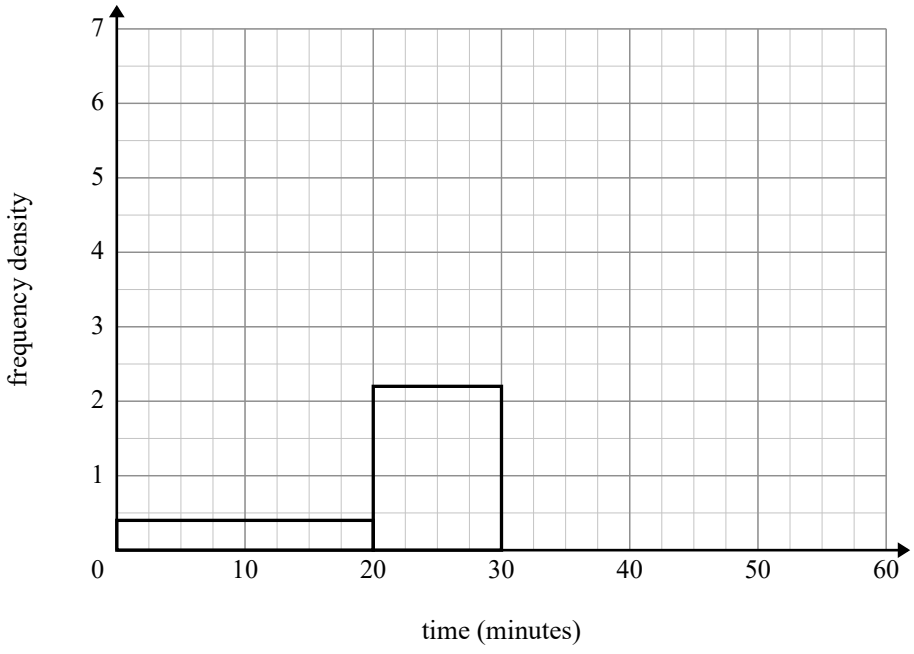
**22** The table shows the times, $t$ minutes, taken by 120 students to finish a test.

| Time ($t$ minutes) | $0 < t \leqslant 20$ | $20 < t \leqslant 30$ | $30 < t \leqslant 35$ | $35 < t \leqslant 40$ | $40 < t \leqslant 60$ |
|---|---|---|---|---|---|
| Frequency | 8 | 22 | 30 | 34 | 26 |

**(a)** Calculate an estimate of the mean time.

..... min [4]

**(b)** Complete the histogram to show this information.
The first two bars have been drawn for you.

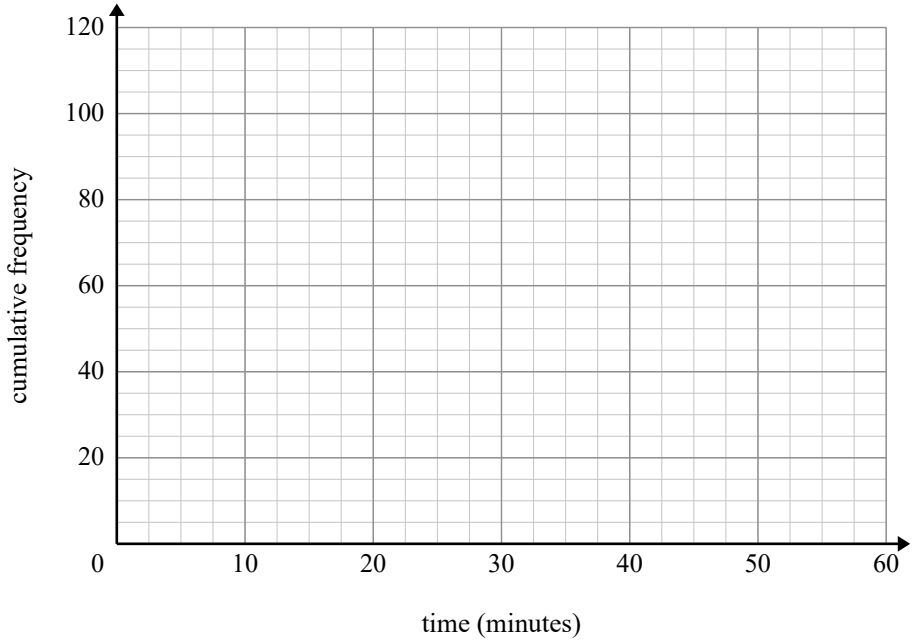



[3]

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**(c)** Complete the cumulative frequency table and draw the cumulative frequency diagram on the grid.

| Time ($t$ minutes) | $t \leqslant 20$ | $t \leqslant 30$ | $t \leqslant 35$ | $t \leqslant 40$ | $t \leqslant 60$ |
|---|---|---|---|---|---|
| Cumulative frequency | 8 | | | | 120 |




[2]

**(d)** Use your diagram to estimate the interquartile range.

..... min [2]

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